M9.3.b.1 Semantic Web and Ontologies

Transversal homework assignment

The objective of this work is to design and validate a complete semantic knowledge base in a domain of your choice based on the courses and labs developed the last 4 weeks. Your work must demonstrate: OWL reasoning, Rule-based reasoning with SWRL, Constraint validation with SHACL, ontology design principles, and as far as possible, the use of ontology design patterns.

The work must be carried out by **groups of 2 students**. For each of the steps 1 to 6 hereafter, report and discuss the choices, the processes, the obtained results, and illustrative relevant screenshots in **one document to be submitted** on Celene together with a **list of versions of the ontology,** 1 version per question (2 to 6) **in ttl format**. You should be prepared to answer questions and discuss your work during the presentation session.

The use of AI assistants is allowed but should be commented all along the reported document. You should clearly state which parts are generated and which ones are your own creation. **Note that the AI generated content should not exeed 50% of the final assignement.**

1- Start by choosing a domain to model in a similar to way to university in last labs. Ensure that the shosen domain is suitable to be modelled throw classes, relationships, constraints, and rules.

2- Model an ontology that includes:

a. Classes and properties:

- At least 8–10 classes organized using
  - subclass hierarchies
  - disjointness
  - domain and range constraints
- At least 8–10 object and data properties
- At least one property chain

b. Individuals

- Create at least 2 instances per class representing realistic situation in the domain of your ontology.

3- OWL Reasoning: Apply a DL reasoner to demonstrate possible inferences of class subsumption, properties, instance classification, as well as possible inconsistencies. In the latter case, correct the ontology to make it consistent.

4- Rule-based Reasoning:
Create **at least 4 SWRL rules**, including individual classification, new class definition, property propagation, etc.

Run the rule-based reasoner and show the inferred knowledge. You may discuss potential conflicts with OWL semantics and decidability.

5- SHACL constraints:
Create a SHACL file containing **at least 5 shapes** including NodeShape, PropertyShape, Logical constraints, built-in constraints, etc.
Apply the constraints and provide:

- a SHACL validation report
- a discussion of potential violations
- the modifications made to make your ontology compliant with the expressed constraints

6- Ontology engineering best practices:
Note that it is possible to start with this section.
Based on the last course topic on Ontology Engineering best practices,

- check if your ontology complies with the following:
  - reuse of existing vocabularies (foaf, schema.org, dbo, etc…)
  - modularity and separation of concerns
  - correct use of is-a, instance-of, part-of, sub-class-of, etc.
  - naming conventions

- Write the main competency questions in natural language then express them in SPARQL. Run the SPARQL query to check if the ontology correctly answers the competency questions.

- Identify 1 or 2 ontology design patterns from literature that can be integrated into your ontology and show how you apply it/them.